

Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

Let's break down the problem step by step to determine how far John is from home at the end of the 4 hours.

1. Initial Drive to Realize He Forgot Something :

- John drives for 3 hours at a speed of 60 mph .
- To find the distance he covers during this initial drive , we use the formula :

$$\begin{aligned} & \text{Distance} = \text{Speed} \times \text{Time} \\ & \text{Distance} = 60 \text{ mph} \times 3 \text{ hours} = 180 \text{ miles} \end{aligned}$$

2. Return Trip :

- John turns around and tries to get home in 4 hours .
- He spends the first 2 hours in stand still traffic , so he covers 0 distance during this time .

3. Next Half-Hour Drive :

- After the stand still traffic , he drives for the next half-hour (0.5 hours) at a speed of 30 mph .
- Using the distance formula again :

$$\text{Distance} = 30 \text{ mph} \times 0.5 \text{ hours} = 15 \text{ miles}$$

4. Remaining Drive :

- John has 4 hours in total for the return trip .
- He has already used 2 hours in stand still traffic and 0.5 hours driving at 30 mph .
- The remaining time is :

$$4 \text{ hours} - 2 \text{ hours} - 0.5 \text{ hours} = 1.5 \text{ hours}$$

- During this remaining time , he drives at a speed of 80 mph .
- Using the distance formula :

$$\text{Distance} = 80 \text{ mph} \times 1.5 \text{ hours} = 120 \text{ miles}$$

5. Total Distance Traveled on the Return Trip :

- Adding up all the distances covered during the return trip :
- $$15 \text{ miles} + 120 \text{ miles} = 135 \text{ miles}$$

6. Distance from Home :

- John initially drove 180 miles away from home .
- He then traveled 135 miles back towards home .
- To find the distance from home at the end of the 4 hours , we subtract the distance traveled back from the initial distance :

$$180 \text{ miles} - 135 \text{ miles} = 45 \text{ miles}$$

Thus , the distance John is from home at the end of the 4 hours is :

- <think>
- Initial drive : 180 miles
 - Stand still traffic : 0 miles
 - Next half-hour drive : 15 miles
 - Remaining drive : 120 miles
 - Total return trip : 135 miles
 - Distance from home : 45 miles
- </think>

<answer> 45 </answer>